Lecture 8

Hopf algebras and quantum groups

Objectives. *Lecture 7 found braidings by solving equations. Where do they come from? For groups, the tensor product of representations comes from the diagonal $g \mapsto g \otimes g$, duals from $g \mapsto g^{-1}$, and the symmetry from the flip. This lecture asks which algebras H give a monoidal, rigid and braided* $\mathrm{Rep}(H)$*, and proves that bialgebras, Hopf algebras and quasitriangular Hopf algebras suffice. The quantum group $U_q(\mathfrak{sl}_2)$ then produces a braiding that is not a symmetry, and its matrix on two spins is, after a change of gauge, the braid limit of* Integrability*'s XXZ R-matrix. This is a side branch: Lecture 9 needs only that matrix.*

By the end of this lecture you will be able to:

- Define algebras, coalgebras, bialgebras and Hopf algebras in pictures, and prove that a bialgebra makes $\mathrm{Rep}(H)$ monoidal and an antipode makes it rigid.
- Prove that a universal R-matrix gives a braiding $c = P \circ R$, and tell R from c.
- State the reconstruction theorem, and show that Fibonacci and Ising are not representation categories of Hopf algebras.
- Compute with $U_q(\mathfrak{sl}_2)$ on $V_1 \otimes V_1$: the matrix of the braiding, the Hecke relation, and the dictionary to XXZ.

8.1 Bialgebras give monoidal representation categories

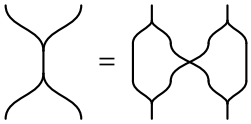
All modules here are finite-dimensional. The algebras acting on them may be infinite-dimensional, as for $U(\mathfrak{g})$ and U_q. The tensor product of linear maps is the ordinary one. An *algebra* is (H, μ, ι), with multiplication $\mu : H \otimes H \to H$ and unit $\iota : \mathbb{C} \to H$, associative and unital. A *coalgebra* is the same with arrows reversed: $(H, \Delta, \varepsilon_H)$, $\Delta : H \to H \otimes H$ coassociative and $\varepsilon_H : H \to \mathbb{C}$ a counit, $(\varepsilon_H \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon_H)\Delta$. In pictures, read upward, μ and Δ are a fusion and a splitting vertex:

$\mu =$  $\Delta =$ 

Kassel writes η for the unit and plain ε for the counit; here η and ε are taken.

Definition 8.1 (Bialgebra)**.** A *bialgebra* is an algebra and a coalgebra on the same H such that Δ and ε_H are algebra maps: $\Delta(xy) = \Delta(x)\Delta(y)$, $\Delta(1) = 1 \otimes 1$, $\varepsilon_H(xy) = \varepsilon_H(x)\varepsilon_H(y)$, $\varepsilon_H(1) = 1$.

The first condition is the *bialgebra law*; with P the flip of Vect (a plain crossing, Lecture 7) it reads

 (8.1)

that is, $\Delta \circ \mu = (\mu \otimes \mu) \circ (\mathrm{id} \otimes P \otimes \mathrm{id}) \circ (\Delta \otimes \Delta)$.

Theorem 8.2 (Bialgebras and monoidal categories)**.** *Let H be an algebra with algebra maps $\Delta\colon H \to H \otimes H$ and $\varepsilon_H\colon H \to \mathbb{C}$. If $(H, \Delta, \varepsilon_H)$ is a bialgebra, then* Rep(H) *is monoidal, with h acting on $V \otimes W$ by $\Delta(h)$ and on $\mathbb{C}$ by $\varepsilon_H(h)$, the associator and unitors those of* Vect*, and the forgetful functor to* Vect *strict monoidal. Conversely · sketched, for finite-dimensional H, if this recipe makes* Rep(H) *monoidal with these associators and unitors, then Δ is coassociative and ε_H a counit.*

Proof. Δ is an algebra map, so $\Delta(h)$ defines an action on $V \otimes W$; likewise ε_H on $\mathbb{C}$. If f and g intertwine, so does $f \otimes g$, since $(f \otimes g)\Delta(h) = \Delta(h)(f \otimes g)$ term by term. The associator of Vect, $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$, intertwines the actions $(\Delta \otimes \mathrm{id})\Delta(h)$ and $(\mathrm{id} \otimes \Delta)\Delta(h)$, which coassociativity makes equal; the unitors intertwine by the counit axiom. The pentagon and triangle hold because they hold in Vect. For the converse, apply the condition to $V = W = H$ itself, acting on itself, and evaluate at $1 \otimes 1 \otimes 1$ (Kassel Prop. XI.3.1). ∎

Example 8.3 (Three bialgebras)**.**

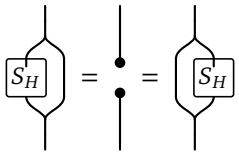
- The group algebra $\mathbb{C}[G]$ with $\Delta(g) = g \otimes g$, $\varepsilon_H(g) = 1$: g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$, and Rep$(\mathbb{C}[G])$ is Rep(G) with its tensor product.
- The functions $\mathbb{C}^G$, with basis δ_g, pointwise product and $\Delta(\delta_g) = \sum_{hk=g} \delta_h \otimes \delta_k$: a module is a G-graded vector space, and Rep$(\mathbb{C}^G)$ is Vec_G; here $\varepsilon_H(\delta_g) = \delta_{g,1}$.
- The enveloping algebra $U(\mathfrak{g})$ with $\Delta(x) = x \otimes 1 + 1 \otimes x$ and $\varepsilon_H(x) = 0$ for $x \in \mathfrak{g}$: this is *QM-I*'s total angular momentum, $J_a = J_a^{(1)} \otimes 1 + 1 \otimes J_a^{(2)}$ (eq. (13.1)), adding angular momenta. (This J is angular momentum, not the structure map of a monoidal functor in Lecture 3.)

The coproduct is the rule for combining two systems. The diagonal $g \otimes g$ says a symmetry acts on both; the primitive $x \otimes 1 + 1 \otimes x$ says a generator, an infinitesimal symmetry, adds.

8.2 Antipodes give duals

Definition 8.4 (Hopf algebra)**.** A *Hopf algebra* is a bialgebra with an invertible linear map $S_H\colon H \to H$, the *antipode*, such that $\mu(S_H \otimes \mathrm{id})\Delta = \iota\,\varepsilon_H = \mu(\mathrm{id} \otimes S_H)\Delta$.

In pictures, with the counit a strand that ends and the unit one that begins:



(8.2)

Proposition 8.5 (Antipodes give duals · *proved given Kassel Thm. III.3.4*)**.** *If H is a Hopf algebra and V an H-module, then V^* with $(h \cdot f)(v) = f(S_H(h)v)$ is an H-module, and the evaluation and coevaluation of* Vect *are H-linear: a left dual. With S_H^{-1} in place of S_H, the same space is a right dual ${}^*V$. So* Rep(H) *is rigid.*

Proof. S_H reverses products, $S_H(xy) = S_H(y)S_H(x)$ (Kassel Thm. III.3.4), so the formula is an action. For $\mathrm{ev}(f \otimes v) = f(v)$, write $\Delta(h) = \sum h_{(1)} \otimes h_{(2)}$; then $\mathrm{ev}(\Delta(h)(f \otimes v)) = \sum f(S_H(h_{(1)})h_{(2)}v) = \varepsilon_H(h)f(v)$, by the antipode axiom: h acts on $\mathbb{C}$ by $\varepsilon_H(h)$, as it should. The coevaluation is checked in the same way with the other half of the axiom (Kassel

Props. III.5.2–5.3). The right dual uses S_H^{-1}, which satisfies the antipode identities for the opposite products, and is checked the same way with the evaluation $V \otimes {}^*V \to \mathbb{C}$ (EGNO Cor. 5.3.7; this is where invertibility is used). ∎

For $\mathbb{C}[G]$, $S_H(g) = g^{-1}$, and V^* is the contragredient representation of Lecture 4. For $U(\mathfrak{g})$, $S_H(x) = -x$: x acts on V^* by $-x^T$, which is why the dual of a spin has the opposite weights. For $\mathbb{C}^G$, $S_H(\delta_g) = \delta_{g^{-1}}$, and the dual of a vector of degree g has degree g^{-1}, as in Vec_G.

8.3 Quasitriangularity gives a braiding

The flip $P\colon v \otimes w \mapsto w \otimes v$ is H-linear for $\mathbb{C}[G]$ and $U(\mathfrak{g})$, because their coproducts are symmetric. In general P intertwines Δ with the opposite coproduct $\Delta^{\mathrm{op}} = P\Delta$, and a braiding needs a correction.

Definition 8.6 (Quasitriangular Hopf algebra)**.** A *universal R-matrix* is an invertible $R \in H \otimes H$ with

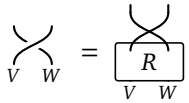
$$R\,\Delta(x)\,R^{-1} = \Delta^{\mathrm{op}}(x) \quad (x \in H),$$
$$(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23},$$
$$(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12},$$

where $R_{12} = R \otimes 1$, and so on. Kassel calls (H, R) *braided* (Def. VIII.2.2).

Theorem 8.7 (R gives a braiding)**.** *For a universal R-matrix, $c_{V,W} = P \circ R_{V,W}\colon V \otimes W \to W \otimes V$ is a braiding on* $\mathrm{Rep}(H)$*, and R satisfies*

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

In pictures, the crossing is a box followed by the plain flip:



(8.3)

Proof. c is H-linear: $c\,\Delta(x) = P\,R\,\Delta(x) = P\,\Delta^{\mathrm{op}}(x)\,R = \Delta(x)\,P\,R$, since $P\Delta^{\mathrm{op}}(x) = \Delta(x)P$. It is natural because R acts through the actions, which intertwiners commute with. For the second hexagon of Definition 7.2, R acts on $(U \otimes V) \otimes W$ by $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$, and $P_{U\otimes V,W} = (P_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes P_{V,W})$; moving $\mathrm{id} \otimes P_{V,W}$ past R_{13} turns it into R_{12}, so $c_{U\otimes V,W} = (c_{U,W} \otimes \mathrm{id})(\mathrm{id} \otimes c_{V,W})$. The first follows in the same way from $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$. Finally the braid relation of Theorem 7.4 for $c = P \circ R$ is, after moving the flips to one side, the displayed equation (Kassel Prop. VIII.3.1, Thm. VIII.2.4). ∎

So R is not the braiding: $c = P \circ R$, Kassel's $\tau \circ R$. This is the *Integrability* equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, now with no spectral parameter, and c satisfies the braid form of Lecture 7. For $\mathbb{C}[G]$ and $U(\mathfrak{g})$, $R = 1 \otimes 1$ and $c = P$: symmetric. But the same Hopf algebra can carry another R. On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$,

$$R = \tfrac{1}{2}\,(1 \otimes 1 + 1 \otimes g + g \otimes 1 - g \otimes g)$$

satisfies the axioms and acts on $V_a \otimes V_b$, where g acts by $(-1)^a$, by $(-1)^{ab}$: the braiding of sVec (Lecture 7). A braiding is structure, and here it is the choice of R.

The algebraic maps have distinct jobs in the representation category. The formulas above give the following dictionary.

algebraic data	operation on modules
μ, ι	Specify the unital and associative laws respected by a module action.
Δ	Define the action on a tensor product of two modules.
ε_H	Define the action on the tensor unit $\mathbb{C}$.
S_H, S_H^{-1}	Define the actions on left and right duals.
R	Correct the flip to the braiding $c = P \circ R$.

Coassociativity makes the ordinary associator a module map. The antipode and universal R-matrix add duality and braiding to this tensor product; neither replaces the coproduct.

8.4 Reconstruction, and what it forbids

A *fibre functor* is a faithful $\mathbb{C}$-linear monoidal functor to Vect, like the forgetful functor of $\mathrm{Rep}(H)$ (EGNO Def. 5.1.1; on a semisimple category the exactness EGNO asks for is automatic).

Theorem 8.8 (Reconstruction · *cited*)**.** *A fusion category with a fibre functor is equivalent to* $\mathrm{Rep}(H)$ *for a finite-dimensional semisimple Hopf algebra* H*, built from the endomorphisms of the functor (EGNO Thm. 5.3.12, applied to fusion categories).*

A fibre functor preserves Frobenius–Perron dimensions (EGNO Prop. 4.5.7), and in Vect they are dimensions, integers. Fibonacci has $\mathrm{FPdim}(\tau) = \varphi$ and Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$, so neither has a fibre functor: no Hopf algebra has Fibonacci or Ising as its representations. Integer FPdims are necessary, not sufficient: Vec_G^{ω} with ω not a coboundary has every FPdim 1 and no fibre functor (EGNO Ex. 5.1.3), since a fibre functor would trivialise ω. The obstruction is FPdim, not the pivotal d: Vec_G for $G = \mathbb{Z}_3$ has the fibre functor that forgets the grading, yet a pivotal structure with $d = e^{2\pi i/3}$ (Lecture 4). The Fibonacci category does come from a quantum group, but as a subcategory of a quotient of its representations at a root of unity, the subject of Lecture 9.

8.5 $U_q(\mathfrak{sl}_2)$ and the XXZ dictionary

Fix $q \in \mathbb{C}^\times$ that is not a root of unity. The algebra $U_q(\mathfrak{sl}_2)$ is generated by E, F, $K^{\pm 1}$ with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F,$$
$$EF - FE = \frac{K - K^{-1}}{q - q^{-1}},$$

and it is a Hopf algebra with

$$\Delta(E) = K \otimes E + E \otimes 1,$$
$$\Delta(F) = 1 \otimes F + F \otimes K^{-1},$$
$$\Delta(K) = K \otimes K,$$
$$S_H(E) = -K^{-1}E, \quad S_H(F) = -FK,$$
$$S_H(K) = K^{-1},$$

and $\varepsilon_H(E) = \varepsilon_H(F) = 0$, $\varepsilon_H(K) = 1$ · *cited* (Kassel Prop. VII.1.1). This Δ is the opposite of Kassel's and EGNO's (Thm. 5.6.2), and S_H is the inverse of Kassel's antipode, as it must be for the opposite coproduct; the course uses it throughout. We use only the modules $V_1^{\otimes n}$ and their direct summands: on them K acts by integer powers of q, and the braiding below exists. (On the one-dimensional module with $K = -1$ and $E = F = 0$, which also satisfies the relations, no braiding is compatible with V_1.) On $V_1 = \mathbb{C}^2$, with basis $e_1 = \uparrow$, $e_2 = \downarrow$,

$$K = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix},$$

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

As $q \to 1$, E, F and $(K - K^{-1})/(q - q^{-1})$ become the e, f, h of $\mathfrak{sl}_2$ (the E, F, H of *Symmetries*, L12), and Δ becomes the primitive coproduct. This is a limit, not substitution at $q = 1$: the presentation above has denominator $q - q^{-1}$.

The construction of the universal R-matrix uses a formal parameter. The ring $\mathbb{C}[[\hbar]]$ consists of formal power series $\sum_{n \geq 0} a_n \hbar^n$, with $a_n \in \mathbb{C}$. Here $\hbar$ is an indeterminate: no numerical value or convergence is assumed. In particular, $q = e^{\hbar/2}$ means its formal exponential series. The numerical parameter q fixed above serves a different role. On the two-spin space the sum involving powers of E and F truncates. The Cartan factor gives diagonal powers of a chosen root $A^2 = q$, yielding the finite matrix below. We use that matrix directly, rather than substituting a complex number into an arbitrary formal power series.

Theorem 8.9 (The braiding on two spins · *cited; proved on* $V_1 \otimes V_1$)**.** *The formal quantum group $U_\hbar(\mathfrak{sl}_2)$, over $\mathbb{C}[[\hbar]]$ with $q = e^{\hbar/2}$, has a universal R-matrix (Kassel Thm. XVII.4.2, his h), and with its legs exchanged it serves our Δ (his Lemma VIII.2.3); it acts on the modules $V_1^{\otimes n}$ and their summands, giving a braiding there (EGNO Ex. 8.3.13). On $V_1 \otimes V_1$, with $q = A^2$, it is $R = \operatorname{diag}(A, A^{-1}, A^{-1}, A)\,(1 + (q - q^{-1})F \otimes E)$, and the braiding is $c = P \circ R = A^{-1} B_q$, with*

$$B_q = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & q - q^{-1} & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}$$

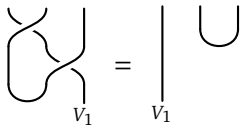
in the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$.

Proof on $V_1 \otimes V_1$. Kassel's series, legs exchanged, is $q^{h \otimes h/2} \sum_n c_n F^n \otimes E^n$ with $c_0 = 1$, $c_1 = q - q^{-1}$; on V_1, $E^2 = F^2 = 0$, so only $n \leq 1$ survives, and $q^{h \otimes h/2} = \operatorname{diag}(A, A^{-1}, A^{-1}, A)$ since $h = \operatorname{diag}(1, -1)$. Direct computation then gives $R\,\Delta(x)R^{-1} = \Delta^{\mathrm{op}}(x)$ for $x = E, F, K$, the two coproduct conditions on $V_1^{\otimes 3}$, and $PR = A^{-1}B_q$ (`scripts/check-l8-algebra.py`). ∎

B_q satisfies the braid relation and the *Hecke relation* $(B_q - q)(B_q + q^{-1}) = 0$: its eigenvalues are q on the triplet and $-q^{-1}$ on the *q-singlet* $v = e_1 \otimes e_2 - q\, e_2 \otimes e_1$, the cup of Lecture 9. So c has eigenvalues A and $-A^{-3}$, and $c^2 \neq 1$: a braiding that is not a symmetry. At $q = 1$, $B_q = P$. The q-singlet is invariant: $\Delta(E)v = \Delta(F)v = 0$ and $\Delta(K)v = v$, so $v \colon \mathbb{C} \to V_1 \otimes V_1$ is a module map, and V_1 is its own dual. The operator $U = B_q - q$, the projection onto the singlet up to a factor, satisfies $U^2 = -(q + q^{-1})U$ (Sheet 8); it is the generator of the Temperley–Lieb algebra of Lecture 9.

Naturality of the braiding with the cup $v \colon \mathbb{C} \to V_1 \otimes V_1$ demands that a strand pass the

cup as if it were not there, $c_1c_2(v \otimes \mathrm{id}) = \mathrm{id} \otimes v$:



(8.4)

Write $e_{ijk} = e_i \otimes e_j \otimes e_k$. Applying the displayed matrix B_q on the adjacent factors gives

$$\begin{aligned} B_1B_2(v \otimes e_1) &= qe_{112} - q^2e_{121} \\ &= q(e_1 \otimes v), \\ B_1B_2(v \otimes e_2) &= qe_{212} - q^2e_{221} \\ &= q(e_2 \otimes v). \end{aligned}$$

Thus for a scalar $z \in \mathbb{C}^\times$ and a crossing $c = zB_q$, cup naturality requires $z^2q = 1$. Choosing a root A with $A^2 = q$ gives $z = A^{-1}$; the other sign corresponds to the other root. The Hecke-normalised B_q solves the braid relation, but it fails this naturality condition.

The following XXZ comparison is optional context for readers of *Integrability*. No later argument uses the spectral parameter. That course's eq. (15.22) gives a trigonometric R-matrix with spectral parameter λ and anisotropy $\Delta_{\mathrm{XXZ}} = \cos\eta$. The subscript distinguishes this number from the coproduct map Δ. Its spectral limits give B_q and its inverse:

Integrability	this course
anisotropy $\Delta_{\mathrm{XXZ}} = \cos\eta$	$q = e^{i\eta} = A^2$, $\Delta_{\mathrm{XXZ}} = (q + q^{-1})/2$
$R(\lambda)$, after the gauge $\mathrm{diag}(e^{\lambda/2}, e^{-\lambda/2})$ on factor 1	the same, $\tilde{R}(\lambda)$
$\lim_{\lambda\to-\infty}(-2e^{\lambda})P\tilde{R}(\lambda)$	B_q, the over-crossing up to A
$\lim_{\lambda\to+\infty}(2e^{-\lambda})P\tilde{R}(\lambda)$	B_q^{-1}, the under-crossing up to A^{-1}
$\eta \to 0$	$q \to 1$, $B_q \to P$

Sheet 8 derives the displayed spectral limits, including the gauge change and scalar factors. The two limits give the two crossings, with the categorical normalisation $c = A^{-1}B_q$ fixed above.

Summary.

- *Handed on:* a coproduct combines two systems; a bialgebra makes $\mathrm{Rep}(H)$ monoidal (Theorem 8.2), an antipode makes it rigid (Proposition 8.5), a universal R-matrix makes it braided, with $c = P \circ R$ (Theorem 8.7).
- *Reconstruction:* a fibre functor makes a fusion category $\mathrm{Rep}(H)$ (Theorem 8.8); integer FPdims are necessary, so Fibonacci and Ising are not $\mathrm{Rep}(H)$.
- *The quantum group:* on two spins, $c = A^{-1}B_q$ with $(B_q - q)(B_q + q^{-1}) = 0$ (Theorem 8.9); B_q is, after a change of gauge, the braid limit of the XXZ R-matrix.
- *Next:* Lecture 9 turns B_q into the Temperley–Lieb and Kauffman-bracket calculus, adds the twist, and builds $\mathrm{SU}(2)_k$.

References

Kassel, *Quantum Groups*: algebras, coalgebras, bialgebras and antipodes (Thm. III.3.4), §§III.1–III.3, pp. 39–52; modules and their duals, Props. III.5.1–III.5.3, pp. 58–60; $U(\mathfrak{g})$,

Prop. V.2.4, p. 97; $U_q(\mathfrak{sl}_2)$, Def. VI.1.1, p. 122, and its Hopf structure, §VII.1, p. 140; braided bialgebras, Def. VIII.2.2, Lemma VIII.2.3 and Thm. VIII.2.4, pp. 173–175, and their braidings, Prop. VIII.3.1, p. 178; Prop. XI.3.1, p. 285; the universal R, Thm. XVII.4.2, p. 413. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fibre functors, Def. 5.1.1, p. 91; reconstruction, Thm. 5.3.12, p. 96; duals, Cor. 5.3.7, p. 95; $U_q(\mathfrak{sl}_2)$, §5.6, p. 101; quasitriangular Hopf algebras, Def. 8.3.1, p. 198; the $U_q(\mathfrak{sl}_2)$ braiding, Ex. 8.3.13, p. 200; FPdim preserved, Prop. 4.5.7, p. 73. Heunen and Vicary, *Categories for Quantum Theory*: monoids and comonoids, Defs. 4.1, 4.3, pp. 128–129; Hopf algebras, Defs. 6.20, 6.31, pp. 205–210. Majid, *A Quantum Groups Primer*, Defs. 1.1–1.3 and 5.1, pp. 2–4, 29.